Learning Objective 8.3: I know the forms of Maxwell's equations for time-varying fields and use them to calculate electric and magnetic fields for a variety of physical systems.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **A pickup coil.** A square loop 20 cm on a side lies in the xy plane in a uniform field $\mathbf{B} = \hat{\mathbf{z}}\, 0.5 \sin(120\pi t)$ mT.

   (a) Find the amplitude of the voltage induced around the loop.

   amplitude = ☐

   (b) We replace the loop with a coil of 50 turns of the same size. Find the new amplitude.

   amplitude = ☐

   (c) Just after $t = 0$, B is positive and increasing. Seen from $+z$, which way does the induced current flow in the single loop? Explain with Lenz's law.

2. **Checking a pair of fields.** A lossless, source-free region has $\epsilon_r = 4$ and $\mu_r = 1$. A proposed electric field is $\mathbf{E} = \hat{\mathbf{y}}\, 10\cos\left(2\pi \times 10^9\, t - \beta z\right)$ V/m.

   (a) Use Faraday's law to find the magnetic field **H** that must accompany **E**, in terms of β.

   (b) Use the Ampere–Maxwell law to find the value of β for which the pair satisfies Maxwell's equations, and the wavelength.

   β =

   λ =

   (c) Find the amplitude of **H** and its direction.

   H_0 =

   (d) Show that the pair also satisfies both of Gauss's laws.

3. **Conservation of charge.** In a region, the current density is $\mathbf{J} = \hat{\mathbf{r}}\, kr$ A/m^2 with $k = 2$ A/m^3, in spherical coordinates.

   (a) Find the rate of change of the charge density, $\partial \rho_v / \partial t$.

   $\partial \rho_v / \partial t =$ [ ]

   (b) Find the current out of a sphere of radius 10 cm, and show that it equals the rate at which the charge inside decreases.

   current = [ ]

   (c) Explain why Ampere's law without the displacement current could not describe this region.

4. **An ideal transformer.** A transformer runs from a 240 V rms, 50 Hz line with $N_1 = 960$ turns on the primary and supplies 24 V rms to an $8\ \Omega$ load. The core has a cross-sectional area of 10 cm^2.

   (a) Find the number of secondary turns and both currents.

   N_2 = ____

   I_1 = ____

   (b) Find the peak flux in the core and the peak flux density.

   B_{pk} = ____

   (c) A student proposes saving copper by using $N_1 = 240$ and $N_2 = 24$, the same ratio. Explain what happens to the core.

Documentation: _____